typst-doctorate-template

Author's guide and reference

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github.com/suvendubrk/typst-doctorate-template

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Preface

This guide describes `typst-doctorate-template`, a two-sided book template for doctoral theses written in Typst. It was made for a thesis in theoretical physics and therefore cares about mathematics: equation numbering per chapter and per appendix, displays that break and number as in LaTeX, running text that keeps its line spacing around notation, and macros for Dirac notation.

The guide is itself set with the template, so its pages show the result of every setting it describes. Part I explains how to write a thesis with it; Part II explains how it is built, for those who want to change it. To start your own thesis, keep the structure of the folder, delete the guide chapters and `chapters/_guide.typ`, and write your chapters in their place.

PART I

Using the Template

◇ Chapter 1 ◇

Getting started

IN THIS CHAPTER

1.1 The files of a thesis 1
1.2 Building the thesis 1
1.3 Fonts 2
1.4 Writing a chapter 2

This document is both the manual of the template and an example of its output: every page you read was set by it. Each command is shown as source code in a grey box, followed by the result it produces. The last chapter, Chapter 7, explains how the template is built, so that you can change it.

The template is written for Typst, a typesetting system with a markup language close to Markdown and a scripting language for everything else. No package is needed: the whole template is the single file `template/thesis.typ`.

1.1 The files of a thesis

A thesis is split over several files, so that each chapter can be edited and previewed on its own.

File or folder	Purpose
`main.typ`	The whole thesis. Applies the template once and includes every other file. Render this.
`preview.typ`	Renders a single file with the full styling, for fast previews of one chapter.
`meta.typ`	Shared settings and macros. Every chapter imports it, so all files share one configuration (`conf`).
`template/thesis.typ`	The template itself: layout, numbering, commands.
`template/aps-doi.csl`	Citation style: American Physical Society, with DOI links.
`frontmatter/`	Title pages and preface.
`chapters/`	One file per chapter. The guide chapters and `_guide.typ` can be deleted.
`backmatter/`	Coauthor contributions, summaries, acknowledgements.
`references.bib`	The bibliography database (BibTeX format).
`fonts/`	Optional fonts not built into Typst, such as KpMath and STIX Two.

1.2 Building the thesis

From the folder of the thesis, render the whole document with

```
typst compile --font-path fonts --ignore-system-fonts main.typ
typst watch   --font-path fonts --ignore-system-fonts main.typ   # rebuild on save
```

`--font-path fonts` makes Typst find the fonts in `fonts/`, and `--ignore-system-fonts` guarantees that the result does not depend on what happens to be installed on your computer. The script `build.py` runs the same command from Python. In VS Code, the Tinymist extension previews the thesis live; the settings in `.vscode/` point it at the same fonts.

A long thesis takes several seconds to render. To work on one chapter, render `preview.typ` instead and tell it which file to show:

```
typst watch --font-path fonts preview.typ --input file=chapters/ch4_maths.typ
```

References to other chapters cannot be resolved in such a preview, so they are shown as small yellow labels instead of failing.

1.3 Fonts

The template works with the fonts built into Typst alone. The settings in `meta.typ` prefer two free fonts when they are present in `fonts/`:

- **KpMath** (Kepler) for mathematics, available from CTAN
- **STIX Two** for headings and calligraphic letters, available from GitHub.

When a font is missing, Typst falls back to the next name in the list and prints a warning. Any other font can be used by changing `body-font`, `heading-font` and `math-font` (Section 3.1).

1.4 Writing a chapter

A chapter is a file that imports `meta.typ` and starts with a level-one heading. `main.typ` includes it inside `#mainmatter[...]`.

```
#import "../meta.typ": *

= Spin chains <ch:spins>

In this chapter we study ...

== The Heisenberg model <sec:spins-heisenberg>

$ H = J sum_j bold(S)_j dot bold(S)_(j+1) $ <eq:spins-heisenberg>
```

Give every chapter, section, equation and figure that you cite a label, written in angle brackets, and refer to it with `@`. Prefixing labels with their kind and chapter (`ch:`, `sec:`, `eq:`, `fig:`, `tab:`, `app:`) keeps them unique across a long thesis.

The source of this guide is a good starting point: each command described here appears in the chapter files exactly as shown.

◇ Chapter 2 ◇

Document structure

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A thesis is a book: it has front matter, a main body divided into parts and chapters, and back matter. This chapter shows how each level is written. Most of these commands change the whole document, so they are shown as source only; their effect is visible in this PDF itself.

2.1 Front, main and back matter

`main.typ` wraps the thesis in three blocks. Each sets the page numbering and heading style of everything inside it.

```
#include "frontmatter/titlepage.typ"

#frontmatter[            // roman folios i, ii, ...; unnumbered headings
  #tableofcontents()
  #include "frontmatter/preface.typ"
]

#mainmatter[             // arabic folios from 1; chapters numbered 1, 2, ...
  #include "chapters/ch1_intro.typ"
]

#backmatter[             // unnumbered headings; side tabs and header stop
  #include "backmatter/summary.typ"
  #bibliography("references.bib", title: "Bibliography",
                style: "template/aps-doi.csl")
]
```

Inside the front and back matter, write headings as `#heading(numbering: none) [Preface]` or simply `= Preface`; they are unnumbered automatically, still appear in the table of contents, and each gets a side tab of its own. The tabs start at the table of contents, so title pages and dedications have none.

2.2 Parts

A part divider is a full page with the part number and title, placed on a right-hand page. Its pages are not counted, so the page numbering of the chapters runs on without a gap. The part appears as a centred row in the table of contents.

```
#part[Integrable Open Systems]
#part("Time-Dependent Problems", subtitle: "Exact solutions",
      color: rgb("#46543d"), font: "Libertinus Serif")
```

2.3 Chapters and sections

Every level-one heading inside the main matter opens a chapter on a new page, with a "Chapter N" kicker, the title, a dashed rule and a mini table of contents ("In this chapter"). Lower levels are numbered sections:

```
= Chapter title <ch:label>          // Chapter 3
== Section <sec:label>              // 3.1
=== Subsection                      // 3.1.1
==== Sub-subsection                 // 3.1.1.1
```

The mini table of contents is controlled globally by the option `chapter-minitoc` and per chapter by a line placed just before the heading:

```
#minitoc-next(false)   // no mini-TOC for the next chapter only
= Introduction
```

The mini-TOC can also be placed by hand anywhere with `#minitoc()`, for example with a different title: `#minitoc(title: "Overview")`.

2.4 Per-chapter appendices

Long derivations can be moved to appendices that belong to a chapter. They are lettered A, B, … within the chapter, open after the chapter's last page, and number their contents with the letter: sections "2.A.1", equations "(2.A.1)", figures "Figure 2.A.1". The side tab stays on the parent chapter.

```
= Some chapter
... text ...
#chapter-appendix("Detailed derivation", id: <app:deriv>)[
  == Preliminaries                  // 2.A.1
  $ R(u) = 1 + u P $ <eq:deriv-r>   // (2.A.1)
]
```

The `id` makes the appendix citable: `@app:deriv` renders "Appendix 2.A". This chapter ends with such an appendix, Appendix 2.A, which contains Equation (2.A.1).

2.5 Tables of contents

```
#tableofcontents()                        // all levels
#tableofcontents(title: [Contents], depth: 2)
#list-of-figures-and-tables()             // one combined list
```

The look of the entries is set in the template (Section 7.7).

2.6 Chapter references

Typst calls every heading a “Section” in references. `main.typ` therefore contains one line that makes references to chapters read “Chapter N”:

```
#show heading.where(level: 1): set heading(supplement: [Chapter])
```

Appendix 2.A ◇ A demonstration appendix

An appendix is written like a chapter. Its equations carry the appendix letter,

$$T(u) = R_{0N}(u)\cdots R_{01}(u), \tag{2.A.1}$$

and Equation (2.A.1) can be cited from anywhere in the thesis.

2.A.1 A section of the appendix

Sections inside an appendix are numbered with the letter as well.

◇ Chapter 3 ◇

Configuration

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All visual settings are arguments of the function `thesis()`. They are collected once, in `meta.typ`, as `conf`, which `main.typ` and `preview.typ` both apply. Change a setting there and every file follows.

```
// meta.typ
#let conf = thesis.with(
  title: "Thesis title",
  author: "Your Name",
  font-size: 10pt,
  body-font: "Libertinus Serif",
  math-font: ("KpMath", "New Computer Modern Math"),
  heading-font: ("STIX Two Text", "Libertinus Serif"),
  page-width: 170mm, page-height: 240mm,
  margin: (inside: 24mm, outside: 20mm, top: 24mm, bottom: 22mm),
  theme: "black",
  link-color: rgb("#9F3D1A"),
)

// main.typ
#show: conf
```

3.1 Fonts and page

A font argument takes one name or a list of names; the first available one is used. The page size is either a named paper (`paper: "a4"`, `"iso-b5"`, …) or an explicit trim, which wins when both `page-width` and `page-height` are given. Printers usually specify a trim and margins; copy them here. `inside` is the margin at the binding.

3.2 Colours

The accent colour (chapter kickers, rules, tabs, mini-TOC) comes from a named theme or is given directly. Links (citations, references, URLs) use `link-color`, which defaults to the accent.

Theme	Accent
`black`	`#4e4e4e`
`classic`	`#1f3a5f`
`modern`	`#073929`

The side tabs run from the table of contents to the bibliography: every unnumbered section of the front and back matter gets its own tab (labelled i, ii, … and I, II, …), between

which the chapters sit. When there are more tabs than fit the page edge, they shrink. Giving the front and back matter colours that continue the chapter palette makes the edge of the printed book read as one band.

New themes are added to the dictionary `themes` in the template (Section 7.7). With `tab-colors`, each chapter gets its own tab colour; the list is cycled when there are more chapters than colours. The highlight boxes, framed equations and `btable` rules of a chapter follow its tab colour (Chapter 5).

3.3 All options

Option	Default	Meaning
Metadata		
`title`, `author`	`none`	Written into the PDF properties.
`lang`	`"en"`	Language, for hyphenation and quotes.
Fonts		
`body-font`	Libertinus Serif	Running text.
`heading-font`	Libertinus Serif	Headings, tabs, running header.
`math-font`	New Computer Modern Math	Equations.
`font-size`	`9.5pt`	Body text size.
Page		
`paper`	`"iso-b5"`	Named paper size.
`page-width`, `page-height`	`auto`	Explicit trim; overrides `paper` when both are set.
`margin`	inside 26, outside 20, top 24, bottom 22 mm	Page margins of a two-sided book.
Colours		
`theme`	`"classic"`	Key into `themes`.
`accent`	`none`	Explicit accent colour; overrides `theme`.
`part-color`	`#464737`	Part rows in the table of contents.
`link-color`	`auto`	Citations and references; `auto` is the accent.
Chapter opener		
`chapter-minitoc`	`true`	Automatic “In this chapter” list.

Option	Default	Meaning
`chapter-raise`	`-1.1cm`	Shift of the chapter title towards the page top.
Side tabs		
`tab-side`	`"outer"`	`"outer"`, `"right"` or `"left"`.
`tab-numbering`	`"1"`	Label style: `"1"`, `"A"`, `"a"`, `"I"`, `"i"`.
`tab-label-size`	`9pt`	Size of the label on the tab.
`tab-active-fill`	`none`	Tab colour; `none` uses the accent or `tab-colors`.
`tab-label-color`	`white`	Label colour.
`tab-colors`	`none`	One colour per chapter, cycled.
`tab-numbering-front`	`"i"`	Labels of the front-matter tabs.
`tab-numbering-back`	`"I"`	Labels of the back-matter tabs.
`tab-colors-front`, `tab-colors-back`	`none`	Colours of the front- and back-matter tabs; `none` continues the `tab-colors` cycle.
Equations		
`eq-per-chapter`	`true`	Number per chapter, (2.4), and per appendix, (2.A.4).
`eq-numbering`	`"(1.1)"`	Format: first symbol chapter, second equation.
`eq-numbering-appendix`	`auto`	Appendix format; `auto` adds a lettered level.
`eq-break`	`false`	`true`: every numbered display may split across pages, number once beside the last line.
`eq-break-height`	`9cm`	With `eq-break: false`, numbered displays shorter than this never split.
`eq-num-gap`	`1em`	Least space between a formula and its number.
Inline maths		
`inline-flush`	`9em`	Inline formulas up to this width keep the line spacing even.
`inline-reach`	top 0.32em, bottom 0.38em	How far they may reach before the line opens up.
`inline-compact`	`true`	Delimiters in running text keep text size.
Figures and tables		

Option	Default	Meaning
`fig-per-chapter`	`true`	Number per chapter, "Figure 2.3".
`fig-numbering`	`"1.1"`	Format of figure and table numbers.
`fig-numbering-appendix`	`auto`	Appendix format.
Flip-book		
`flipbook`	`none`	A function `n => content` drawn in the page corner (Section 5.5).
`flipbook-size`	`18mm`	Size of the corner cell.
`flipbook-guide`	`false`	Outline the cell on every page.

◇ Chapter 4 ◇

Mathematics

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Typst writes mathematics between dollar signs. A formula with a space after the opening and before the closing dollar is a display; without, it sits in the text. This chapter covers what the template adds: numbering, page breaking, the behaviour of inline formulas, and a set of macros for physics.

4.1 Display equations

Displays are numbered per chapter. A label after the closing dollar makes the equation citable.

```
The Schrödinger equation,
$ i planck partial_t ket(psi) = hat(H) ket(psi) , $ <eq:maths-schrodinger>
is cited as @eq:maths-schrodinger.
```

The Schrödinger equation,

$$i\hbar\partial_t|\psi\rangle = \hat{H}|\psi\rangle, \tag{4.1}$$

is cited as Equation (4.1).

Lines are separated by \ and aligned at &. A multi-line display carries one number.

```
$
  hat(H) &= sum_j (hat(S)^x_j hat(S)^x_(j+1) + hat(S)^y_j hat(S)^y_(j+1)) \
         &= 1/2 sum_j (hat(S)^+_j hat(S)^-_(j+1) + hat(S)^-_j hat(S)^+_(j+1)) .
$
```

$$\begin{aligned} \hat{H} &= \sum_j \left(\hat{S}^x_j \hat{S}^x_{j+1} + \hat{S}^y_j \hat{S}^y_{j+1} \right) \\ &= \frac{1}{2} \sum_j \left(\hat{S}^+_j \hat{S}^-_{j+1} + \hat{S}^-_j \hat{S}^+_{j+1} \right). \end{aligned} \tag{4.2}$$

To leave out numbers, wrap displays in `nonumeq`; to frame one, use `boxeq`.

```
#nonumeq[$ e^(i pi) + 1 = 0 $]
#boxeq[$ [hat(x), hat(p)] = i planck $]
```

$$e^{i\pi} + 1 = 0$$

$$[\hat{x}, \hat{p}] = i\hbar \tag{4.3}$$

4.1.1 Wide formulas

Typst centres a display and pins its number to the right margin. When the two would come closer than `eq-num-gap`, the template moves the formula to the left; if even that leaves no room, the number drops to its own line below the formula, as in LaTeX.

```
$ Z = sum_(n=0)^infinity (-1)^n binom(2n, n) x^n y^(2n) exp(-beta E_n) cos(n theta) $
```

$$Z = \sum_{n=0}^{\infty} (-1)^n \binom{2n}{n} x^n y^{2n} \exp(-\beta E_n) \cos(n\theta) \tag{4.4}$$

4.1.2 Page breaks

By default, a numbered display of ordinary height is never split between two pages; it moves to the next page whole, because Typst would otherwise print its number on both pieces. Only displays taller than `eq-break-height` may split. To let one particular display split at its line breaks, wrap it in `breakeq`. Its number is then drawn once, next to its last line.

The choice can also be made for the whole document: with `eq-break: true` in `conf`, every numbered display may split in this way, and `nobreakeq` keeps a single one whole.

```
#breakeq[$
  f(x) &= a_0 + a_1 x + a_2 x^2 + a_3 x^3 \
       &quad + a_4 x^4 + a_5 x^5 + a_6 x^6 + dots
$ <eq:maths-long>]
#nobreakeq[$
  g(x) &= b_0 + b_1 x \
       &quad + b_2 x^2 + dots
$]
```

$$\begin{aligned} f(x) &= a_0 + a_1 x + a_2 x^2 + a_3 x^3 \\ &\quad + a_4 x^4 + a_5 x^5 + a_6 x^6 + \dots \end{aligned} \tag{4.5}$$

$$\begin{aligned} g(x) &= b_0 + b_1 x \\ &\quad + b_2 x^2 + \dots \end{aligned} \tag{4.6}$$

4.2 Inline formulas

Formulas in running text are kept on the line grid: a subscript, a ket or a small fraction does not push its line away from the next, so a paragraph full of notation keeps even spacing. Only a formula that reaches far above or below the line, such as a stacked fraction with scripts, opens up its own line by the necessary amount (`inline-reach`).

```
A state $ket(psi_n)$ with energy $E_n = planck omega (n + 1/2)$ and
amplitude $braket(phi, psi)$ keeps the line spacing of the text, while
$display(sum_(k=1)^N 1/k^2)$ makes room for itself.
```

A state $|\psi_n\rangle$ with energy $E_n = \hbar\omega(n + \frac{1}{2})$ and amplitude $\langle\varphi|\psi\rangle$ keeps the line spacing of the text, while $\displaystyle\sum_{k=1}^{N} \frac{1}{k^2}$ makes room for itself.

In running text, brackets keep the size of the text instead of stretching around their content (`inline-compact`). To make one bracket enclose its content in full, give it an explicit size:

```
Compare $(sqrt(binom(2s, q)))^N$ with $lr(size: #100%, (sqrt(binom(2s, q))))^N$.
```

Compare $(\sqrt{\binom{2s}{q}})^N$ with $\left(\sqrt{\binom{2s}{q}}\right)^N$.

4.3 Notation for physics

```
$ ket(psi), quad bra(phi), quad braket(phi, psi), quad
  braket(phi, hat(O), psi), quad expval(hat(O)), quad
  dket(rho), quad dbra(rho), quad bigO(t^(-1)) $
```

$$|\psi\rangle, \quad \langle\varphi|, \quad \langle\varphi|\psi\rangle, \quad \left\langle\varphi\middle|\hat{O}\middle|\psi\right\rangle, \quad \left\langle\hat{O}\right\rangle, \quad |\rho\rangle\rangle, \quad \langle\langle\rho|, \quad \mathrm{O}(t^{-1}) \tag{4.7}$$

Macro	Result
`ket(x)`, `bra(x)`	Dirac kets and bras, with bars and angles of matching height.
`braket(a, b)`, `braket(a, O, b)`	Overlap and matrix element.
`expval(x)`	Expectation value $\langle x\rangle$.
`dket(x)`, `dbra(x)`	Vectorised states, $\lvert\rho\rangle\rangle$.
`bigO(x)`	Order symbol with an upright O.
`cal(A)`	Calligraphic letters from STIX Two Math, $\mathcal{ALR}$.
`ox`, `oplus`, `oxbig`, `oplusbig`	Tensor product and direct sum, $a \otimes b$, $a \oplus b$.

These live in `template/thesis.typ` (Dirac notation) and `meta.typ` (project macros). New macros belong in `meta.typ`, where every chapter sees them (Section 7.8).

4.4 Sums and products

In a display, Typst stacks the limits of a sum above and below the sign. A chapter that starts with `#show: compact-sums` sets them beside the sign instead, which keeps dense equations flat. Single sums can still be stacked with `limits()`.

```
#show: compact-sums
$ sum_(j=1)^N hat(S)^z_j quad "vs." quad
limits(sum)_(j=1)^N hat(S)^z_j $
```

$$\sum_{j=1}^{N} \hat{S}_j^z \quad \text{vs.} \quad \sum_{j=1}^{N} \hat{S}_j^z \tag{4.8}$$

A long condition under a sum is split over lines with `substack`. When it is still wider than the sign, `smashop` lets it hang under the neighbouring terms instead of pushing them away (LaTeX's `\smashoperator`). For a product, pass `op: $limits(product)$`.

```
$ x + limits(sum)_(substack(i < j, i\, j in {alpha^((q))})) l_(i j)
  quad "vs." quad
  x + #smashop($limits(sum)_(substack(i < j, i\, j in {alpha^((q))}))$) l_(i j) $
```

$$x + \sum_{\substack{i<j \\ i,j \in \{\alpha^{(q)}\}}} l_{ij} \quad \text{vs.} \quad x + \sum_{\substack{i<j \\ i,j \in \{\alpha^{(q)}\}}} l_{ij} \tag{4.9}$$

Matched braces and bars normally stretch around their content, so a set with a superscript such as $\{\alpha^{(q)}\}$ grows taller than its surroundings. A chapter with `#show: flush-sets` draws braces around simple content, and the bars of a set's size, at their natural height. Braces around fractions or big operators still stretch.

```
$ limits(sum)_(i in {alpha^((q))})
abs({beta^((p))}) $
#[#show: flush-sets
$ limits(sum)_(i in {alpha^((q))})
abs({beta^((p))}) $]
```

$$\sum_{i \in \{\alpha^{(q)}\}} \left| \left\{ \beta^{(p)} \right\} \right| \tag{4.10}$$

$$\sum_{i \in \{\alpha^{(q)}\}} |\{\beta^{(p)}\}| \tag{4.11}$$

4.5 Fine spacing

Macro	Width	Use
`neg`	$-1/6$ em	Negative thin space, TeX's `\!`.
`neg2`	$-1/3$ em	Larger negative space.
`tightx`	-0.08 em	Hair-line tightening.
`wide`	0.6 em	Space between a formula and its condition.

◇ Chapter 5 ◇

Figures, tables and boxes

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5.4 Placeholders 18
5.5 Flip-book 18

5.1 Figures

Figures are numbered per chapter (“Figure 5.1”) and float to the top of a page by default, as in LaTeX. Images can be SVG, PNG, JPEG or PDF.

```
#figure(
  image("../images/ch5/spectrum.svg", width: 80%),
  caption: [*Spectrum of the model.* Bold text opens the caption.],
) <fig:floats-spectrum>
```

A figure that must stay where it is written takes `placement: none`. Captions are set small, justified and left aligned; Section 7.7 shows where to change that.

```
#figure(
  rect(width: 60%, height: 1.6cm, fill: luma(90%)),
  caption: [A figure kept in place.],
  placement: none,
) <fig:floats-demo>
See @fig:floats-demo.
```


Figure 5.1: A figure kept in place.

See Figure 5.1.

Several panels share one number and caption with `subfigure`. The panels are labelled (a), (b), … and cannot be cited one by one.

```
#subfigure(
  rect(width: 100%, height: 1.2cm, fill: luma(88%)),
  rect(width: 100%, height: 1.2cm, fill: luma(80%)),
  caption: [Two related panels.],
)
```


(a) **(b)**

Figure 5.2: Two related panels.

5.2 Tables

`btable` draws a table in the style of LaTeX's `booktabs`: no vertical rules, heavy rules at the top and bottom, a light rule under the header. Rows can be tinted with `highlight` (1 is the first body row), and `inset` narrows the cells of a wide table. Rules and tints take the tab colour of the chapter (or the accent when `tab-colors` is not set).

```
#figure(
  btable(
    columns: 3,
    header: ([Model], [Symmetry], [Method]),
    highlight: (2,),
    [XXX], [$frak(s u)(2)$], [Bethe ansatz],
    [XXZ], [$U_q (frak(s u)(2))$], [Bethe ansatz],
    [Ising], [$bb(Z)_2$], [Free fermions],
  ),
  caption: [Some integrable spin chains.],
) <tab:floats-chains>
```

Model	Symmetry	Method
XXX	$\mathfrak{su}(2)$	Bethe ansatz
XXZ	$U_q(\mathfrak{su}(2))$	Bethe ansatz
Ising	$\mathbb{Z}_2$	Free fermions

Table 5.1: Some integrable spin chains.

Tables in a figure are numbered "Table 5.1" independently of figures. Unlike images, they stay in place and may break across pages, repeating their header row.

5.3 Boxes

`concept` (alias `highlight-box`) sets a key idea in a box tinted with a light shade of the chapter's tab colour, with a bar in the colour itself; without `tab-colors` the accent is used, and `color:` sets any other. `boxeq` follows the same colour. Equations inside keep the full width, so their numbers align with the others.

```
#concept(title: "Main result")[
  The model is integrable for every value of the coupling,
  $ [hat(H), hat(Q)_n] = 0 . $
]
#concept(color: rgb("#7b2d8b"))[A box in another colour.]
```

Main result

The model is integrable for every value of the coupling,

$$\left[\hat{H}, \hat{Q}_n\right] = 0. \tag{5.1}$$

A box in another colour.

5.4 Placeholders

While a figure is missing, `placeholder-fig` (from `meta.typ`) reserves its space:

```
#figure(placeholder-fig(height: 1.5cm), caption: [To be drawn.], placement: none)
```

figure placeholder

Figure 5.3: To be drawn.

5.5 Flip-book

A thesis can carry a small animation in the corner of its pages, visible when the book is flipped through. `flipbook` takes a function from the page number to the content of the corner cell. `meta.typ` contains a moving dot as a demonstration:

```
// meta.typ
#let demo-flip(n) = {
  let t = calc.rem(n, 16) / 16
  let x = 100% * calc.abs(2 * t - 1)
  box(width: 100%, height: 100%,
    place(left + horizon, dx: x, circle(radius: 2.5pt, fill: rgb("#5A6C80"))))
}
```

```
#let conf = thesis.with(..., flipbook: demo-flip)

// frames from image files instead:
//   flipbook: n => image("images/frames/" + str(n) + ".png")
```

◇ Chapter 6 ◇

References and back matter

IN THIS CHAPTER

6.1 Cross-references 20
6.2 Citations and bibliography 20
6.3 Coauthor contributions 20
6.4 Title pages and summaries 21

6.1 Cross-references

A reference is written `@label` and renders as a link with the kind and number of its target. Numbers are always computed at the target, so a reference to an equation of another chapter, or of an appendix, shows the right number from anywhere.

```
@ch:maths, @sec:maths-sums, @eq:maths-schrodinger,
@eq:structure-demo, @app:structure-demo, @fig:floats-demo,
@tab:floats-chains.
```

> Chapter 4, Section 4.4, Equation (4.1), Equation (2.A.1), Appendix 2.A, Figure 5.1, Table 5.1.

The word in front of the number can be replaced for one reference with a supplement in brackets, `@eq:maths-long[Eq.]` giving Equation (4.5), or dropped by referring to the number alone with `#ref(<eq:maths-long>, supplement: none)`.

6.2 Citations and bibliography

Citations use the same syntax with a key of `references.bib`:

```
Factorised scattering @yang-1967-article-deltainteraction and the
reflection equation @sklyanin-1988-article-boundary; see also the
monograph by Baxter @baxter-1982-book-exactlysolved.
```

> Factorised scattering [1] and the reflection equation [2]; see also the monograph by Baxter [3].

The bibliography is printed where `#bibliography(...)` stands in `main.typ`, in the style given there. `template/aps-doi.csl` follows the American Physical Society and adds a DOI link to each entry. Any other Citation Style Language file can be used, or a built-in style such as `"ieee"` or `"nature"`.

6.3 Coauthor contributions

Theses built on published papers usually need a page listing each paper and the share of each author. `backmatter/contributions.typ` defines a small command for it, with the journal linked through the DOI and the preprint through its arXiv number:

```
#let me = text(weight: "bold")[YN]   // your initials, in bold

#entry(
  [@ch:spins: Spin chains],
  [*Y. Name*, A. Coauthor and B. Supervisor],
  [Physical Review B 100, 000000 (2026)],
  doi: "10.1103/xxxx", arxiv: "2601.00000",
)[#me performed the calculations and wrote the text. ...]
```

6.4 Title pages and summaries

`frontmatter/titlepage.typ` holds the title page and its reverse (copyright, printer, funding). Universities often supply an official title page that must be used unchanged; place it there. The summaries and acknowledgements in `backmatter/` are ordinary files with an unnumbered heading.

PART II

Changing the Template

How the template works

IN THIS CHAPTER

7.1 Typst in five ideas 22
7.2 The three layers 23
7.3 A map of thesis.typ 23
7.4 Shared state 23
7.5 Numbering 24
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7.7 Recipes 25
7.8 Adding your own macros 26

This chapter is for readers who want to change the template rather than use it. It assumes the basics of Typst, which are summarised first, and then walks through `template/thesis.typ`.

7.1 Typst in five ideas

Set rules change the default value of an element's property from that point on, within the current scope:

```
#set text(size: 10pt)
#set heading(numbering: "1.1")
```

Show rules replace how an element is drawn. The function receives the element (`it`) and returns new content. A show rule can also apply a set rule to one kind of element only:

```
#show heading.where(level: 2): it => text(fill: blue, it)
#show figure.caption: set text(size: 0.8em)
```

Context gives access to where content ends up: the page it falls on, the value of a counter there, or the current font size. Everything that depends on the layout is written inside `context { ... }`.

Counters and states carry values through the document. `counter(math.equation)` numbers equations; `state("name", initial)` holds any value and is changed with `.update(...)` at a position in the document and read with `.get()` (here) or `.at(location)` (elsewhere).

Queries find elements anywhere in the document, `query(heading.where(level: 1))`, and `measure(content)` returns the size content would have. Typst lays the document out several times until such values stop changing.

7.2 The three layers

Layer	Role
`template/thesis.typ`	Generic: knows nothing about a particular thesis. Defines `thesis()` and all commands.
`meta.typ`	Project: imports the template, re-exports it, adds project macros, and fixes the configuration `conf`.
chapters	Content: import `meta.typ` and nothing else.

Because every file imports `meta.typ`, a macro or setting added there is available everywhere, and `preview.typ` renders a chapter with exactly the styling of the full build.

7.3 A map of thesis.typ

The file is divided into numbered sections:

Section	Contents
§ 0 Shared state	States through which the commands learn the configuration and the region they are in.
§ 1 Constants and themes	Ink colours of the table of contents and captions, the indent step, the theme palette. Safe to edit.
§ 2 Layout helpers	Numbering helpers, the side tabs, the flip-book cell and the running header.
§ 3 Content helpers	`concept`, `boxeq`, `nonumeq`, `breakeq`, spacing, `subfigure`, `btable`, Dirac notation.
§ 4 Outlines	Table of contents, list of figures and tables, the mini-TOC.
§ 5 Structure	`frontmatter`, `mainmatter`, `backmatter`, `part`, `chapter-appendix`.
§ 6 The template	The function `thesis()`: page set-up, headings, numbering, references, equations.

7.4 Shared state

Commands such as `concept` or `minitoc` are called inside chapters, far from the `thesis()` call that holds the configuration. They read it through states, which `thesis()` sets at the start of the document:

State	Meaning
`_accent`, `_headfont`, `_chapraise`	Accent colour, heading font and title raise, published by `thesis()`.
`_region`	`"chapter"` or `"chap-appendix"`: decides whether numbers carry an appendix letter.
`_apphead`, `_ispart`	True exactly at an appendix heading or a part heading, so the heading rules can tell them from chapters.

State	Meaning
`_mtoc`	The per-chapter mini-TOC override set by `minitoc-next`.
`_pstart`	Page on which a part divider starts, to uncount its pages.
`_eqbreak`	`true` inside `breakeq`, `false` inside `nobreakeq`, `auto` elsewhere (the option `eq-break` decides).
`_tabcolors`	The list `tab-colors`, from which `_chapter-color()` picks the colour of the current chapter for boxes and tables.

Three invisible markers complete the picture: `backmatter()` places one (`<thesis-tabstop>`) where the running header stops, every part-divider page carries another (`<thesis-divpage>`), and every front- or back-matter section that owns a side tab carries a third (`<thesis-tabpos>`, placed by `tableofcontents()` and by unnumbered level-one headings, with the matter read from the state `_matter`). `_tab-entries()` collects these and the chapters, in order, into the list of tabs.

7.5 Numbering

Chapter appendices are level-one headings that must not advance the chapter counter, so they carry `numbering: none` and are recognised through `_apphead`. Inside `chapter-appendix`, the section numbering is replaced by a function that inserts the letter.

The show rule for level-one headings (§ 6) resets the counters of equations, figures and tables at every chapter and appendix, and draws the chapter opener. An equation's number is produced by `_eq-num(location, ...)`, which reads the chapter counter, the appendix letter and the equation counter at that location. The same function is used for the tag beside the equation and in the `show ref` rule, which rewrites every reference to an equation, figure, table or appendix from the target's own location. This is why references never disagree with the numbers they point to.

7.6 Equations

The show rule for numbered displays measures the formula without its number and then decides:

1. The local choice is read first: `breakeq` forces a split, `nobreakeq` keeps the display whole; elsewhere the option `eq-break` decides.
2. A display that may split is drawn without Typst's number, and the number is placed once, beside the last line, whose height is measured from the formula (the last line may sit inside a bracket group spanning several lines).
3. With `eq-break: false`, a display taller than `eq-break-height` is left to Typst, and all others are made unbreakable.

4. If formula and number fit with `eq-num-gap` on both sides, the display is left as Typst draws it; if they fit only with the formula moved left, a grid places them side by side; otherwise the number goes below.

A table has to remain a plain `table` for `figure` to number it as a table, so `btable` cannot read the chapter colour itself, which needs `context`. It draws its rules and tints in placeholder colours instead, and a show rule on `table` in `thesis()` replaces them with shades of the chapter's colour (`_btable-colors`).

Inline formulas are boxed at the height of a capital letter so that they do not change the line spacing (`inline-flush`). Before boxing, the rule measures how far the formula really reaches; only the part beyond `inline-reach` is added back as invisible struts, which open that one line as much as needed.

7.7 Recipes

To change	Edit
A theme colour	Add an entry to `themes` (§ 1) and select it with `theme:`.
Colours of the table of contents or captions	`_toc-ink`, `_toc-ink-2`, `_toc-ink-3`, `_caption-ink` (§ 1).
Indent of the table of contents	`_toc-indent` (§ 1).
Size and shape of the side tabs	The `box(width: 8mm, height: 12mm, ...)` and the spacing in `_side-tabs` (§ 2).
Running header	`_running-header` (§ 2): what recto and verso pages show.
Chapter opener	The `is-chapter` branch of the level-one heading rule (§ 6): kicker, title size, rule.
Section headings	The rules for heading levels 2 to 4 at the start of the heading part of § 6.
Captions	The `show figure.caption` rules in § 6.
Part pages	`part()` (§ 5): background tint, sizes, the "PART I" line.
Table rules	`btable` (§ 3).

Small changes are best made from outside the template, so that it can be updated later. Set and show rules placed in `main.typ` after `#show: conf` override the template for the whole thesis; placed at the top of a chapter, they apply to that chapter only:

```
// main.typ, after #show: conf
#show heading.where(level: 2): set text(fill: rgb("#1f3a5f"))
#set par(leading: 0.7em)
#show figure.caption: set text(size: 0.85em)
```

7.8 Adding your own macros

Project notation goes into `meta.typ`. A macro is a function; written with math content, it is used inside formulas like any built-in.

```
// meta.typ
#let tr = math.op("tr")                    // upright operator: tr A
#let comm(a, b) = $lr([#a, #b])$           // commutator: comm(A, B)
#let dd = math.upright("d")                // upright differential
```

A rule that only some chapters need, like `compact-sums` or `flush-sets`, is written as a function of the whole body and applied with `#show: name` at the top of those chapters:

```
// meta.typ
#let compact-sums(body) = {
  show math.equation: eq => {
    show sym.sum: it => math.scripts(it)
    show sym.product: it => math.scripts(it)
    eq
  }
  body
}

// in a chapter
#show: compact-sums
```

Coauthor contributions

All chapters of this thesis were written under the supervision of …

Chapter 1: Getting started

Y. Name, A. Coauthor and B. Supervisor
Journal Name 1, 100 (2026)
arXiv:0000.00000

YN performed the calculations and wrote the text. AC contributed to the analysis. BS supervised the project.

I

Summary

The English summary of a thesis goes here, bearing the title of the thesis.

Samenvatting

De Nederlandse samenvatting van het proefschrift, met de titel ervan.

III

Acknowledgements

The acknowledgements of a thesis go here.

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